

Q1. Given an ^(+ve element) $arr[N]$, find the max^m value of k such that all subarrays of size k has sum $\leq B$.

$B=20$

ex $\Rightarrow$

0 1 2 3 4 5 6 7
3 2 5 4 6 3 7 2

$\Rightarrow$

$\downarrow$
~~25 > 20~~

$k=1$ ✓

$k=2$ ✓

$k=3$ ✓

$k=4$ ✓

$k=5$ ✗

6 ✗

7 ✗

8 ✗

9 ✗

:

$O(p=4)$

$k=1$ to 8

mid = 7

mid = 2

$k=1$ — N

for($k=1; k \leq N; k++$) {

sliding window of
size k , to check
all subarr.

$O(N^2)$



i) all subarrays of size (mid) have
 $sum \leq B$

$ans = mid$

$l = mid + 1$

ii) any subarrays of size (mid) have
 $sum > B$

$\rightarrow$ go left

$r = mid - 1$

int ans;

$l = 1, r = N$

while ($l \leq r$) {

$mid = \frac{l+r}{2}$

if (check (arr, mid, B)) {

$ans = mid$

$l = mid + 1$

} else {

$r = mid - 1$

}

} return ans;

boolean check(arr, mid, B) {

use sliding window of
size, and check sum of
subarrays $\leq B$

}

$$\boxed{\begin{array}{l} TC \Rightarrow O(N \log N) \\ SC \Rightarrow O(1) \end{array}}$$

Q2. Are magical numbers +

No. of multiple of 4 from $[1, 100] \rightarrow \frac{100}{4} = 25$

No. of multiple of 6 from $[1, 100] \rightarrow \frac{100}{6} = 16$

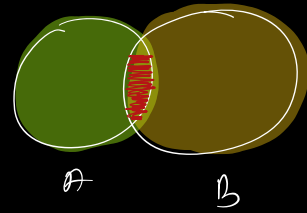
No. of multiple of 4 or 6 from $[1, 100]$

$$\begin{aligned} &\rightarrow \frac{100}{4} + \frac{100}{6} - \frac{100}{\text{LCM}(4,6)} \\ &= \frac{100}{4} + \frac{100}{6} - \frac{100}{12} = 25 + 16 - 8 \\ &= \underline{\underline{33}} \end{aligned} \quad \left| \begin{array}{l} \text{LCM}(4,6) = 12 \\ 12, 24, 36, 48 \\ 60, 72, 84, 96 \end{array} \right.$$

Count of multiples of A or B, from

1 to x

$$\rightarrow \frac{x}{A} + \frac{x}{B} - \frac{x}{\text{LCM}(A, B)}$$



$$A \cup B = A + B - A \cap B$$

Given A & B, find Nth magical no.

↓
multiple of A or B

$$A = 2$$

$$B = 3$$

$$N = \underline{\underline{8}}$$

2, 3, 4, ~~5~~, 6, ~~7~~, 8, 9, 10, ~~11~~, 12
8, ~~13~~, 14

1 2 3 4 5 6 7 8 9

a, b

1 ————— min(a, b) ————— N * (min(a, b))

$$a = 2$$

$$b = 1000$$

$$N = \underline{\underline{5}}$$

2 4 6 8 10 12 14 — — —

→ 5x2 or, 5x min(2, 1000)

```

for (i = min(a, b); i <= N * (min(a, b)); i++) {
    if (i is magical)
        count++
    if (count == N)
        return i
}

```

$O(N * \min(a, b))$

* cf $\rightarrow$ count of factors

ans = min(a, b) □ $N * (\min(a, b))$

$\downarrow$

mid

$$cf = \frac{mid}{a} + \frac{mid}{b} - \frac{mid}{\text{lcm}(a, b)}$$

$$\frac{cf < N}{l = mid + 1}$$

$$\frac{cf == N}{\text{ans} = mid}$$

$$r = mid - 1$$

$$\frac{cf > N}{r = mid - 1}$$

$$a = 5 \quad b = 7 \quad N = 3$$

$$\text{mid} \Rightarrow \text{cf} = \underline{\underline{N}}$$

cf $\Rightarrow$ 1 1 2 2 2 3 3 3 3 4 5

$$N = 4$$

$\min(a, b)$	$N \times (\min(a, b))$		
l	r	mid	check
5	20	12	$\frac{12}{5} + \frac{12}{7} - \frac{12}{35} = 3$
13	20	16	$\frac{16}{5} + \frac{16}{7} - \frac{16}{35} = 5$
13	15	14	$\frac{14}{5} + \frac{14}{7} - \frac{14}{35} = 4 \rightarrow \text{ans (14)}$
13	13	13	$\frac{13}{5} + \frac{13}{7} - \frac{13}{35} = 3$
14	13		
	11		
	stop		

$$\begin{cases} a \times b = \text{lcm}(a, b) \times \text{gcd}(a, b) \\ \text{lcm}(a, b) = \frac{a \times b}{\text{gcd}(a, b)} \end{cases}$$

$$TC \Rightarrow O(\log(N \times \min(a, b)))$$

=

$$TC = \text{gcd}$$

$$\log(\underline{\underline{\min(a, b)}})$$

→ Aggressive Cows

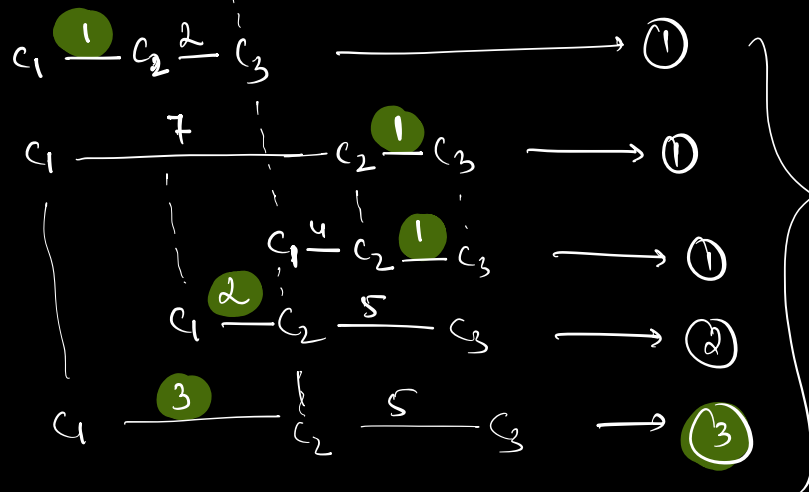
N stalls, M cows ($M \leq N$)

1 stall → 1 cow

maximise the min^m dist between

cows, return dist. $P_0 \rightarrow$ stall locations (asc order)

stall → 1 2 4 8 9 ($M=3$)



$O(P \cdot 2)$

1 2 4 8 9 ($M=3$)
 c_1 c_2

min^m dist → 1 → 1
 max^m dist → $arr[N-1] - arr[0]$ → 8

1 2 4 8 9 (M=3)

$$D = 1 \quad \checkmark \quad C_1 - C_2 - C_3$$
$$D = 2 \quad \checkmark \quad C_1 \xrightarrow{3} C_2 \xrightarrow{4} C_3$$
$$D^23 \quad \checkmark \quad C_1 \xrightarrow{3} C_2 \xrightarrow{4} C_3$$

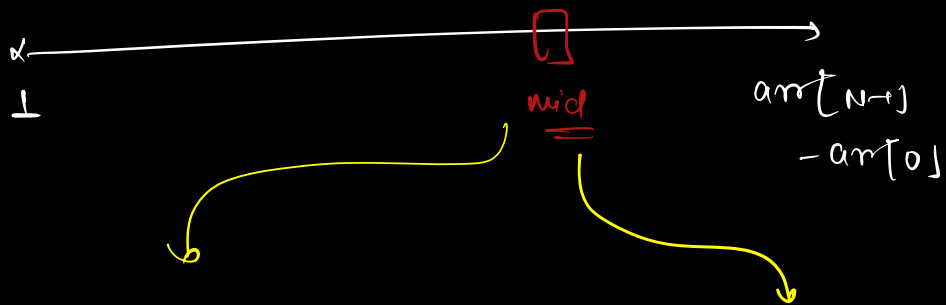
$D=4$ ~~X~~ C_1 ————— C_2 ————— X
~~X~~ ~~X~~ ~~X~~

S X

b x

7 ~~7~~

Ans (range)



we are able
to store all
vars atleast
mid dist apart

Ques = What if
1 = What if 1

if we can't store
all cases ahead—
(mid) dist apart—

← go left
 $r = \text{mid} - 1$

M=4

2 6 11 14 19 25 30 39 43

C₁ ————— C₂ ————— C₃ ————— C₄

l	r	mid	check
1	41	21	X
1	20	10	✓ ans = 10
11	20	15	X
11	14	12	✓ ans = 12
13	14	13	X
13	12		
X			

return ans = 12

l = 1

r = arr[N-1] - arr[0]

while (l <= r) {

mid = l + $\frac{(r-l)}{2}$

if (check(arr, mid, M)) {

ans = mid

l = mid + 1

$\frac{l+r}{2}$

l = INF_MAX - 1

r = INF_MAX - 2

l + r

= 2 (INF_MAX

<pre> } else { r = mid - 1; } } return ans; </pre>	$ \begin{aligned} & l + \frac{(r-l)}{2} \\ & = l + r/2 - l/2 \\ & = l/2 + r/2 \\ & = \frac{l+r}{2} \end{aligned} $
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boolean check(arr, mid, M) {

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    prevCowLoc = arr[0]

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    cow = 1

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    for (i = 1; i < N; i++) {

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        if (arr[i] - prevCowLoc >= mid) {

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```

            prevCowLoc = arr[i]

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```

            cow++;

```

```

        if (cow == M)

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            return true

```

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        }

```

```

    }

```

```

    return false

```

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}

```

$$\begin{array}{l} TC = O(N \log N) \\ SC = O(1) \end{array}$$

2 6 11 14 19 25 30 39 43
 c₁ c₂ c₃ c₄

prev = ~~2~~ 14 25 39

c = ~~1~~ ~~2~~ ~~3~~ 4 = M

Wid = 10

return pre